

*Standard Operating Procedure **Belt Sander***

General International 15-015M1 (Belt/Disc Sander)

The 15-015M1 is a heavy-duty combination sander featuring a **1 HP motor** and a cabinet base, designed for both horizontal and vertical sanding. It is a rough shaping and material removal tool and is not intended for high-precision finished cuts.

Potential Hazards

- Be aware of the following potential hazards when working with Horizontal Band saws:

- **Mechanical**: Skin abrasion from direct exposure to the high-pressure media stream.
- **Respiratory**: Inhalation of hazardous dust, especially when blasting coated materials or using silica-based media.
- **Operational**: Static electricity buildup, reduced visibility due to dust, and foot pedal accidental activation.
- **Energy Sources**: High-pressure compressed air (typically 80–100 psi) and electrical power for the dust collector/lighting.

Personal Protective Equipment (PPE)

PPE Item	Requirement
Eye Protection	**Safety Glasses (ANSI Z87.1):** Required at all times to protect against metal shards and projectile debris.
Footwear	**Closed-Toe Shoes:** Steel-toed boots are recommended when handling heavy plate stock or large sheet material.
Hand Protection	**Strictly Forbidden:** Gloves are a major entanglement hazard with rotating spindles and fluted drill bits



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PPE Item	Requirement
Clothing/Hair	Tie back long hair and secure loose clothing. Remove all jewelry, especially rings, which can be caught in material during the feed.

Before Use

- **Media Inspection**: Ensure the abrasive media is clean and free of moisture or large debris that could clog the blast nozzle.
- **Glove Inspection**: Check the integrated rubber gloves for punctures or thinning. **Never** use the blast gun if your skin is exposed inside the cabinet.
- **Dust Collector**: Ensure the dust collection system/vacuum is turned **ON** before beginning. This maintains visibility and prevents internal pressure buildup.
- **Window Guard**: Verify the protective film on the glass window is clear. Replace if it is too pitted to see the workpiece safely.
- **Air Pressure**: Set the regulator to the manufacturer's recommended pressure (usually not exceeding 100 psi for most cabinet models).

During Use

- **Door Integrity**: Ensure all cabinet doors are latched and sealed before engaging the air blast.
- **Grip**: Hold the blast gun firmly with one hand and the workpiece with the other.
- **Blast Angle**: Avoid blasting directly at the viewing window or the gloves. Aim the stream at a 45–60 degree angle to the workpiece for maximum efficiency.
- **Static Discharge**: Be aware that static shocks can occur; maintain contact between the workpiece and the cabinet grate if possible.
- **Visibility**: If dust obscures your view, stop blasting and allow the dust collector to clear the air before continuing.



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After Use

- **Dwell Time**: Wait 30 seconds after the final blast before opening the doors to allow the dust collector to remove suspended particulates.
- **Part Cleaning**: Use the internal air blow-off gun (if equipped) to remove excess media from the part before taking it out of the cabinet.
- **Media Recovery**: Ensure the grating is clear of large debris so media can fall back into the hopper.
- **Cleanup**: Sweep up any media that may have spilled onto the floor. Blasting media acts like ball bearings on a hard floor and is a significant slip hazard.
- **Hygiene**: Wash hands and forearms thoroughly to remove abrasive dust.

Emergency Equipment and Procedures

- **E-Stop/Air Cutoff**: Know the location of the quick-disconnect for the air line and the power switch for the dust collector.
- **Reporting**: Report any cabinet leaks, glove failures, or visibility issues to the Laboratory Project Supervisor immediately.
- **Reporting**: All blade failures, injuries, or near-misses must be reported to the Laboratory Project Supervisor immediately.